

School of Computer Science

COMP 2067 Programming for Beginners

Summer 2025 – Lab 6

Course instructor: Dr. Olena Syrotkina

Functions

This content is protected and may not be shared, uploaded, or distributed. Any unauthorized distribution, reproduction or sharing of this material, including uploading to CourseHero, Chegg, StuDocu or other websites, is strictly prohibited and may be subject to legal action. Students are granted access to this material solely for their personal use and may not use it for any other purpose without the express written consent of the instructor.

OBJECTIVES:

- i. To use built-in functions and methods available with Python 3
- ii. To create user-defined functions and use them in a program

1. Launch **IDLE** in your computer, open a **new file** through IDLE, and name it **lab6sol.py**.

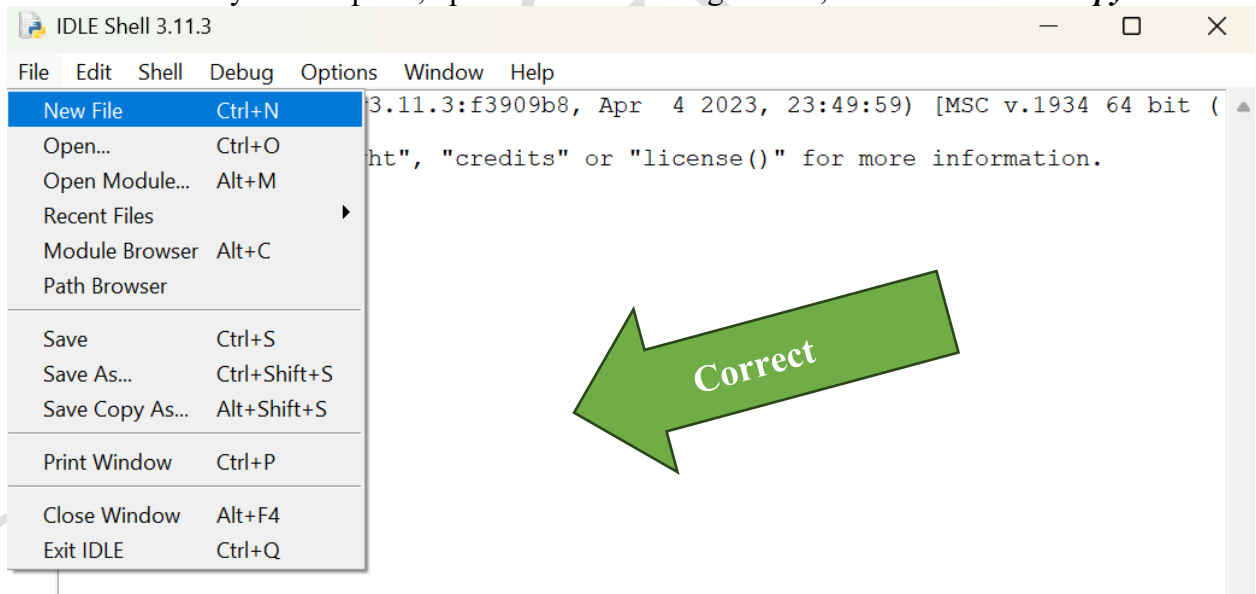


Fig. 1: Creating a new file

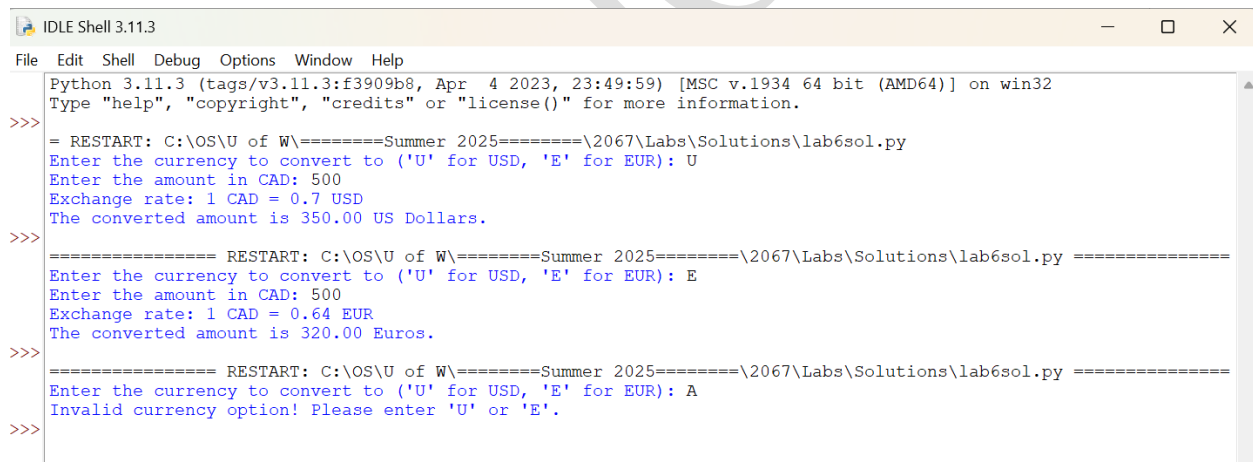
2. In **lab6sol.py**, write a Python program **with comments** to do the following:
 - a. Define a function **convert_currency(amount, currency)** that converts a given amount in CAD to another currency and returns the converted value:
 - i. If **currency == 'U'**, it converts the amount to US Dollars using a fixed exchange rate (e.g., 1 CAD = 0.7 USD).
 - ii. If **currency == 'E'**, it converts the amount to Euros (EUR) using a fixed exchange rate (e.g., 1 CAD = 0.64 EUR).
 - iii. Otherwise, it prints: "Enter the correct currency." and returns None.

- b. Use built-in functions to:
 - i. Remove any extra spaces from input: **strip()**;
 - ii. Convert the input to uppercase: **upper()**;
 - iii. Check if the input is alphabetic **isalpha()**;
 - iv. Validate that it is exactly one character long: **len()**.
 - v. If input is invalid, print a message and stop the program.
 - c. In the main program, ask the user to input which **currency** to convert to. The user should enter 'U' for USD or 'E' for EUR. Assign the entered value to the variable **currency**. Then, ask the user to input the amount in CAD and assign the numerical value to the variable **amount**. Ensure the user inputs a **positive number**.
 - d. Call the **convert_currency()** function with the validated amount and currency. If conversion is successful, print the result with a suitable message.
3. **Submit your Python code** (i.e., as *lab6sol.py* file) through **Brightspace**.

NOTE: Do NOT submit a screenshot or the contents of the IDLE shell displaying your output. These will NOT be marked. You must submit the *lab6sol.py* file.

Below is a **sample** screenshot showing example outputs for Lab 6.

Please note: do not submit a screenshot; only submit your lab6sol.py file. Your suitable message does not need to match the screenshot exactly.



```

IDLE Shell 3.11.3
File Edit Shell Debug Options Window Help
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:\OS\U of W\=====Summer 2025=====\\2067\Labs\Solutions\lab6sol.py
Enter the currency to convert to ('U' for USD, 'E' for EUR): U
Enter the amount in CAD: 500
Exchange rate: 1 CAD = 0.7 USD
The converted amount is 350.00 US Dollars.
>>> ===== RESTART: C:\OS\U of W\=====Summer 2025=====\\2067\Labs\Solutions\lab6sol.py =====
Enter the currency to convert to ('U' for USD, 'E' for EUR): E
Enter the amount in CAD: 500
Exchange rate: 1 CAD = 0.64 EUR
The converted amount is 320.00 Euros.
>>> ===== RESTART: C:\OS\U of W\=====Summer 2025=====\\2067\Labs\Solutions\lab6sol.py =====
Enter the currency to convert to ('U' for USD, 'E' for EUR): A
Invalid currency option! Please enter 'U' or 'E'.
>>>
  
```

Fig. 2. A sample screenshot showing example outputs for Lab 6.

Submission:

1. You will earn a maximum of 100 points (accounts for 2.5%) for submitting your **source code** (*lab6sol.py*) within the specified deadline.
2. **Do not upload your source code to Brightspace as a single zip/rar file.** Such submissions will not be accepted and a mark of **zero** will be assigned.
3. Marks will be **deducted** if a *suitable message and/or comments* are missing.
4. Late submissions will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.

5. Unlimited resubmissions are allowed. But keep in mind that we will consider the last submission. That means that if you resubmit after the deadline, a penalty will be applied, even if you submitted an earlier version on time.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.